

Preparing to Build a Project

<https://gemini.google.com/app/d53e46dad830925b>

User prompt: hey study this completely and tell me what are you understand by this after i will tell you what i really want to be make

Response: Across the three documents (presentation pitch, design thinking workbook, and EPICS Review 0 proposal), the project is titled **"AI-Driven IV Fluid Monitoring and Smart Alert System"**.

Here is a breakdown of the core concept:

1. The Core Problem

- **Air Embolism & Complications:** In resource-constrained environments (like MSME hospitals, 24/7 clinics, or floors with high patient-to-nurse ratios), IV monitoring is done manually or using dumb/reactive systems. Empty IV bags or blocked lines go unnoticed, leading to backflow, venous infiltration, or air embolisms.
- **Alarm Fatigue:** Existing automated alerts are strictly reactive; they trigger loud, non-differentiated ward buzzers *after* the bag is already empty or failing, causing severe noise pollution and nursing fatigue.
- **High Cost of Smart Pumps:** Proprietary commercial smart infusion pumps are far too expensive (\$\$\$) for Tier-2/Tier-3 hospitals, leaving around 85% of IV setups completely unmonitored.

2. The Proposed Solution & Pipeline

Instead of a simple static weight threshold or a cloud-dependent system, the project proposes a **low-cost, retrofit edge-AI system** that anticipates failures in advance:

1. **Sense:** A **1 kg Load Cell** continuously measures the physical weight of the hanging IV bag.
2. **Amplify:** An **HX711 24-bit ADC module** digitizes the load cell signal.
3. **Analyze & Predict (Edge AI):** An **ESP32 microcontroller** ingests the weight time-series data locally and runs a **TinyML model** (built/trained via Edge Impulse) to classify flow states into categories like *Normal*, *Blocked*, or *Rapidly Emptying*. It continuously calculates the **exact time-to-empty** locally with near-zero latency.
4. **Triage & Silent Alerts:** The ESP32 communicates over Wi-Fi via a **Blynk IoT backend / Centralized AI Triage Agent**, routing prioritized, silent push notifications to the on-duty nurse's mobile device rather than triggering loud room sirens.

3. Key Design Thinking & Feasibility Takeaways

- **Hardware Selection:** Opted for a load cell + HX711 over optical drip counting (which suffers from ambient light issues) or ultrasonic sensors (which are too expensive).
- **Main Risk / Assumption:** Handling movement artifacts and vibrations on the IV stand so they don't corrupt the time-series inference.

- **Target Impact:** Retrofit existing standard IV poles at minimal cost, target SDG 3 & 9, cut ward alarm noise by up to 40%, and prevent critical patient complications.

Tell me what you really want to build or modify. Are you looking to develop the actual firmware/schematics, implement the TinyML model, replace components, build the mobile/triage interface, or pivot the system entirely?
